Anh (Andre) Tran

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Education

University of Cincinnati — Expected May 2028
Bachelor of Science in Computer Engineering, Minor in Computer Science — *GPA: 3.77/4.00*

Coursework: Data Structures and Algorithms, Object-Oriented Programming, Database Design & Development, Cybersecurity, Digital Design & Digital Systems, Artificial Intelligence, Context Engineering

Experience

University of Cincinnati Department of Computer Science — 08/2026 - Present
Incoming AI Researcher — *Cincinnati, OH*

Commma - Pace your Code (commma.dev) — 05/2026 - Present
Founder & Lead Engineer — *Cincinnati, OH*

- Building **commma.dev** end-to-end, a full-stack **TypeScript** monorepo capturing per-keystroke telemetry, then dual-publishing the extension to both **VS Code Marketplace** and **Open VSX** from one build (400+ downloads).
- Replaced **BullMQ/Redis** with an idempotent interval scanner after idle polls threatened the free-tier budget, sustaining zero 5xx errors across 33K+ requests.
- Enforced a keys-only privacy guarantee at two independent layers across 271K+ keystroke events, tallying single-character edits client-side and re-deriving privacy mode server-side from **Postgres**.
- Engineered a redundant **GitLab CI/CD** deploy pipeline alongside **GitHub Actions**, restoring full deploy capability within 24 hours during a platform outage, then scaled it back once resolved.

Caphne - Study Buddies Matchmaking (caphne.co) — 01/2026 - Present
Lead Software Engineer — *Remote*

- Cut API response time from 60s to 10s by engineering real-time chat with **Socket.IO** scoped to per-match rooms and a shared **TypeScript** event-name package, sustaining 1.7K+ requests/minute over 30 days in production.
- Led a cross-functional team of 6 engineers through sprint planning, code review, and production deploys in **Nuxt 3**, **Vue**, and **Node.js**, scaling the platform to 400+ active users.
- Architected **AWS** infrastructure (EC2, CloudFront, S3, Route 53) in **Terraform** with **OIDC**-federated credentials, then root-caused and fixed 2 production incidents same-day from raw error output.

KatanaID - AI Branding Toolkit (katanaid.com) — 12/2025 - 05/2026
Founding Engineer — *Remote*

- Bounded latency and API cost across 21 external integrations by leading 5 engineers to architect a **Go** backend fanning out one goroutine per platform per brand-name check, scoped to 4 candidate names per generation.
- Guaranteed zero downtime on missing **Google Gemini** API access and served 2.3K+ requests/day in production by designing a fully deterministic, zero-dependency local fallback.
- Traded reviewed down-migrations for auto-migrate-on-boot velocity by replacing hand-written SQL migrations with **Ent**'s typed schema, eliminating drift across 19+ integrations.

Projects

Draft-Thinker (Live) | *Go, OpenAI API, Qdrant, Redis, Prometheus, Grafana, Docker*

- Cut inference cost 82.9% (bootstrapped 95% CI: 79.6-86.0%) while holding accuracy at 98.1% by designing an entropy router in **Go** using top-k logprobs from the **OpenAI API**.
- Diagnosed why a confidence-threshold baseline plateaus at a 61.5% cost-reduction ceiling from an OpenAI logprob-reporting artifact affecting 65.5% of creative prompts, which entropy routing structurally avoids.
- Built a cancellable speculative-execution pipeline with **Go** goroutines and **context** cancellation, wasting only 5-10% of compute on the 30% of requests that escalate while cutting tail latency.

Ferrox (Live) | *Rust, Systems Programming, Atomics, Mmap, Criterion*

- Built a price-time matching engine in **Rust** with fixed-point prices, trading a 9-22% insert regression for a 67-85% cut in cancel latency via a HashMap-to-BTreeMap swap, benchmarked at 500ns P99 across 4.7M orders/sec.
- Engineered a lock-free SPSC ring buffer with cache-line-padded atomics, choosing Acquire/Release over SeqCst to avoid unnecessary fence instructions, measuring an 8.8x throughput gain over **std::sync::mpsc**.
- Deployed to **AWS Fargate**, measuring the **NLB** at 70% of total hourly cost - more than the compute itself.

Technical Skills

Languages: Go, JavaScript/TypeScript, SQL, Python, Rust, C/C++, C#, Java, HTML/CSS
Technologies: Nuxt.js, Vue.js, Next.js, React, Node.js, Bun, Express.js, Spring Boot, Chi, Gin, Hono, AWS (Amazon Web Services), Docker, PostgreSQL, MongoDB, MySQL, FastAPI, Django, Flask, LangChain/LangGraph, ChromaDB, scikit-learn, NumPy, Socket.IO, Git, GitHub, Grafana, Prometheus, Redis, Terraform, CI/CD, Linux